



Multimodal study of multilevel pulvino-temporal connections: a new piece in the puzzle of lexical retrieval networks

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Advanced methods of imaging and mapping the healthy and lesioned brain have allowed for the identification of the cortical nodes and white matter tracts supporting the dual neurofunctional organization of language networks in a dorsal phonological and a ventral semantic stream. Much less understood are the anatomical correlates of the interaction between the two streams; one hypothesis being that of a subcortically mediated interaction, through crossed cortico-striato-thalamo-cortical and cortico-thalamo-cortical loops. In this regard, the pulvinar is the thalamic subdivision that has most regularly appeared as implicated in the processing of lexical retrieval. However, descriptions of its connections with temporal (language) areas remain scarce.

Here we assess this pulvino-temporal connectivity using a combination of state-of-the-art techniques: white matter stimulation in awake surgery and postoperative diffusion MRI ($n = 4$), virtual dissection from the Human Connectome Project 3 and 7 T datasets ($n = 172$) and operative microscope-assisted post-mortem fibre dissection ($n = 12$).

We demonstrate the presence of four fundamental fibre contingents: (i) the anterior component (Arnold's bundle proper) initially described by Arnold in the 19th century and destined to the anterior temporal lobe; (ii) the optic radiations-like component, which leaves the pulvinar accompanying the optical radiations and reaches the posterior basal temporal cortices; (iii) the lateral component, which crosses the temporal stem orthogonally and reaches the middle temporal gyrus; and (iv) the auditory radiations-like component, which leaves the pulvinar accompanying the auditory radiations to the superomedial aspect of the temporal operculum, just posteriorly to Heschl's gyrus.

Each of those components might correspond to a different level of information processing involved in the lexical retrieval process of picture naming.

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Introduction

Over the past two decades, elucidating the neural basis of language functions has attracted considerable interest in neuroscience. Advanced methods of imaging and mapping the healthy and lesioned brain provided convergent data supporting a global neuro-functional architecture divided into two main streams^{1,2}: the dorsal phonological route and the ventral semantic route. The cortical areas of the two streams are now well identified, as well as their interconnections through cortico-cortical associative white matter pathways.^{3–6} Much less understood is the substrate of the interaction between the two streams. Two main non-exclusive hypotheses can be proposed. First, some cortical areas can play this integrative role as common targets of dorsal and ventral white matter pathways or by a local connection through short U-fibres.⁷ The second hypothesis would be a subcortically mediated interaction, as most of cortical areas are embedded in crossed cortico-striato-thalamo-cortical and/or cortico-thalamo-cortical loops. In this regard, the role of subcortical structures in language networks has not gained the same attention as cortical ones. Nevertheless, there is ample neuropsychological evidence suggesting that subcortical grey matter structures, including the pulvinar, may play a role in language, especially in lexical processing.^{8–11} Indeed, the concept of subcortical aphasia emerged from lesion studies¹² and, in particular, the notion of thalamic aphasia,¹³ characterized by lexico-semantic disturbances. More specifically, it has been repeatedly demonstrated that lesions of the left dominant pulvinar correlate with a fluent type aphasia, characterized by naming deficits, semantic paraphasia and perseverations.¹³ Hence, the pulvinar is thought to be involved in the process of lexical selection. However, despite many recent proposals,^{9–16} the exact mechanisms through which the pulvinar is involved in language networks are far from being fully understood. In general, the pulvinar is a complex structure with multiple nuclei, thought to play a role in several cognitive functions, including visual processing in relation to attention.^{17–20} The pulvinar has also been involved in integrating information from different sensory modalities, like visual and auditory information.^{21,22} Regarding language, Crosson¹⁶ tentatively put forward the hypothesis that the lexical processing supported by the pulvinar can be considered as declarative memory in nature and that mechanistically, the role of the cortico-pulvinar-cortical connections is to maintain a stable lexico-semantic representation during the iterative word-finding process of procedural memory that narrows down lexical choices to successively smaller neighbourhoods of semantically-related lexical items. As the temporal lobe is viewed as an essential support of declarative memory,²³ we would expect connections between the pulvinar and the temporal lobe to play an important role in this process.

Our poor knowledge of the anatomy of pulvino-cortical connections surely constituted an obstacle to the full elucidation of the

role played by the pulvinar in lexical retrieval. While connections between the pulvinar and Broca's area have been previously thoroughly investigated by tractography,²⁴ very little is known about the connections between the pulvinar and the temporal language areas. Although the anatomy of the fasciculus of Arnold, linking the pulvinar to the region of the temporal pole, has been known since 1838,²⁵ its functional role remains entirely *terra incognita*. Moreover, as the temporal pole is not supposed to be an essential node of lexical retrieval in the temporal lobe, it can be inferred that the pulvinar is also connected to more posterior areas, thought to be involved in lexical processing.^{1,26–28} Accordingly, Penfield and Klingler were able to identify, by fibre dissections, connections between the pulvinar and the middle temporal gyrus.²⁹

In this study, we combined several approaches to describe the anatomy of pulvino-temporal connections and to highlight their under-recognized role in language networks and, more specifically, in lexical retrieval during picture naming. First, we present a short series of four insular glioma patients operated on in awake conditions. Using state-of-the-art tractography, we demonstrated that the stimulation sites causing anomia and located in the temporal isthmus were on the trajectory of pulvino-temporal connections. These observations motivated us to thoroughly investigate these connections. To this end, we dissected high-resolution tractograms of healthy subjects of the 3 T and 7 T datasets from the Human Connectome Project (HCP). In addition to the classical Arnold proper fasciculus, three other contingents of pulvino-temporal connections were identified and named according to their trajectory: a lateral component targeting the middle temporal gyrus, an auditory radiations (AR)-like component running in close relationship to the auditory radiations and targeting the superior temporal gyrus, and an optic radiations (OR)-like component running in close relationship to the OR and targeting the fusiform gyrus. In parallel, we performed white matter fibres post-mortem dissection in humans, as a complementary and confirmatory approach. Finally, we discuss how these new results do not only reconcile sparse previous data on pulvino-temporal connections but could also provide a new piece in the puzzle of neural networks implementing lexical retrieval in picture naming.

Materials and methods

Clinical cases

In this report, we present the manifestations induced by electrical stimulation of the temporal isthmus during the surgical resection of a left insular diffuse low-grade glioma in four patients.

All patient's data were obtained from surgical and mapping protocols that were applied as standard of care. All analyses were thus undertaken retrospectively. The patients provided full informed consent to the use of their data for research purposes.

Direct electrical stimulation at 60 Hz was applied to the cortical surface and white matter while the patient performed tasks assisted by a speech therapist. These tasks included counting during repetitive active movement of the right superior limb (dual task), assessing the individual's ability to perform two distinct cognitive and motor tasks simultaneously. Picture naming tests were also conducted, involving the patient verbally identifying and naming objects, facilitating the evaluation of potential language disorders and especially regarding lexical access. Last, the Pyramid and Palm Trees Test (PPTT) was administered—a task displaying one image on the top row and two on the bottom row, requiring the participant to identify which image in the bottom row is semantically related to the item in the top row. These diverse tasks were designed to assess if electrical stimulation transiently disturbed those cognitive and motor functions. Stimulation was applied with a Nimbus iCare light device (Innopsys) delivering constant-current biphasic square pulses (single pulse duration of 500 μ s, total duration of 1000 μ s) with a bipolar stimulation probe (two straight 1.5 mm diameter electrode tips with a pitch of 5 mm, forming a 'Y'), kept in direct contact with brain tissue for a few seconds (typically 3 s). Stimulus intensity was progressively increased in 0.25 mA steps starting from 1 mA until eliciting an observable interference with patient's behaviour and up to a maximum of 3 mA (to limit the risk of seizure).

Patient tractography

Diffusion-weighted imaging (DWI) was acquired 4 months after the surgical procedure. A 3 T Siemens Skyra system (Siemens) with a 64-channel phased-array head coil, was employed to acquire DWI of the whole head with an anterior-posterior acquisition phase. DWI parameters consisted of an acquisition matrix of $108 \times 108 \times 53$ voxels of 2.3 mm^3 with a total field of view of $250 \times 250 \times 132 \text{ mm}^3$ (interslice gap of 2.5 mm). DWI was acquired along 64 directions with a weighting $b = 2000 \text{ s/mm}^2$. Additionally, five non-weighted ($b = 0$) volumes were also acquired. Echo time was set at 95 ms and repetition time at 7700 ms. The whole sequence lasted 9 min 45 s.

T_1 -image and tractograms were obtained, registered in the MNI space and dissected using the same pipeline as described below for healthy subjects. The TractoFlow pipeline was applied, which includes T_1 and DWI preprocessing steps.³⁰ In a nutshell, the DWI is denoised with mppca (Mrtrix3), eddy/topup corrected with FSL, and N4 corrected with advanced normalization tools (ANTs). The T_1 is denoised and non-linearly warped to upsampled $1 \times 1 \times 1$ DWI data based on the $b = 0$ image and reconstructed diffusion tensor imaging (DTI) fractional anisotropy (FA) map, using the multi-source/multi-targets ANTs version.

We located the epicentre of the intraoperative stimulation site in the temporal isthmus onto the postoperative T_1 image registered to the MNI template. To find the exact location, we relied on the landmarks gathered from the operative reports and the snapshots of the intraoperative neuronavigation. We identified anatomical landmarks in axial, sagittal and coronal planes and positioned a spherical region of interest (ROI, diameter of 4 mm) at the stimulation site using MRicroGL (NITRIC Worcester, USA).

Tractography in healthy subjects

In addition to the previously described patients tractography on standard clinical quality data, we investigated the same pathways in a high-quality dataset of healthy subjects. We opted for HCP

consortium datasets, more specifically the test-retest data on a single 3 T system and matching subjects on a single 7 T system. These two datasets allowed us to investigate the reproducibility of the pathways of interest for intra-scanner (3 T test-retest) and inter-scanner (3 T versus 7 T).

An ensemble tractography strategy was used to reconstruct all complex white matter bundles of the pulvino-temporal structural connectivity. First, whole-brain tractography pipeline was performed using TractoFlow,³⁰ without preprocessing as HCP provided already preprocessed data.³¹ Then, fibre orientation distribution functions were estimated with a fixed response function of $(15, 4, 4) \times 10^{-3} \text{ mm}^2/\text{s}$ using all available b -value data. Finally, particle filtering (PFT) anatomically-constrained tractography (ACT)³² was performed using 10 seeds per voxel of the white matter mask obtained from the registered T_1 -weighted image. Second, we used bundle-specific tractography (BST)³³ to track the full white matter extent of the optic radiations, including Meyer's loop,³⁴ as similarly done with the MAGNET approach.³⁵ BST imposes both anatomical and orientational priors to encourage tractography to traverse hard-to-track bottleneck and crossing fibre regions. Finally, the whole-brain and BST tractography reconstructions were concatenated to produce the final diffusion MRI (dMRI) ensemble tractogram to be dissected.

Reproducibility analysis

We used the Dice coefficient to quantify the spatial overlap, but this score is not appropriate in case of bundle's low density (perfect overlap being challenging) or for evaluating smaller bundles, which may appear dissimilar due to the disproportionate impact of outliers/noise (while larger bundles often easily show high overlap). In such cases, it is better to use the bundle adjacency (in mm) to quantify the shortest distance to overlapping regions. A bundle adjacency score $< 2 \text{ mm}$ ($\pm 1 \text{ mm}$ standard deviation (SD)) at the individual level indicates reconstruction fidelity (based on empirical observation). A bundle adjacency score $< 4 \text{ mm}$ ($\pm 1 \text{ mm}$ SD) at the population level indicates a homogenous population. These agreement metrics have been previously described in detail.^{36,37}

Manual virtual dissection of Arnold tracts

Brain cortical and subcortical areas were parcellated by using recon-all command line of freesurfer 7.1 (Athinoula A. Martinos Center for Biomedical Imaging, Charlestown, USA).³⁸ Cortical regions were dilated by 1 mm before being used as cortical endings ROI in the tractogram dissection. The thalamus was further segmented in its subdivisions using segmentThalamicNuclei.sh.³⁹ This segmentation allowed us to define the lateral geniculate nucleus (LGN) and the pulvinar (gathering in a unique ROI all of its subparts). Other relevant ROIs were manually segmented. These included the two following regions (Supplementary Fig. 1):

- (i) the stem of the inferior longitudinal system (ILS), also named the inferior-fronto-occipital fasciculus (IFOF) (Mandonnet et al.⁴⁰). This ROI was defined on a coronal slice passing through the tip of the temporal horn and comprised the external capsule and the superior part of the temporal isthmus, corresponding to the green areas of the RGB FA-coloured map⁴¹; and
- (ii) the stem of Arnold's bundle proper. This ROI was defined on successive coronal slices of RGB map, as a green area located just inferolaterally to the LGN.

Two coronal planes were also defined: one passing through the tip of the temporal horn (C1) and the other through the posterior

insular point (C2). Last, an axial plane was defined, just above the superior insular sulcus (A1).

Ten hemi-tractograms (five left and five right) were then manually dissected by the senior author under MI-brain software (Imeka Medical Imaging, Sherbrooke, Canada).⁴² These virtual dissections enabled us to segregate the fibres of the pulvino-temporal tractogram in four groups. We noticed that the four branches could be defined according to the criteria outlined in Fig. 1. Of note, at the time these dissections were made, the authors were unaware of existing literature on the topic.

Template creation of Arnold tracts

The five HCP subjects used for the 10 virtual dissection scenes described above (five left and five right hemispheres) were warped to symmetric MNI space using non-linear ANTs registration and averaged ROIs in MNI space were created. We observed that in MNI space, the averaged ROIs were similar to individual ones. This allowed the production of an automatic virtual dissection script to extract the four components of the pulvino-temporal connections. This script gave similar bundle reconstruction quality results whether run in native space or MNI space. Hence, we adopted the MNI space strategy, which could then be applied to any subject or any patient once the tractogram is warped to MNI space, without the need to manually perform the virtual dissection. Finally, we executed this script on the 172 participants of the HCP project (including the five participants manually dissected) who had available data at both 3 T and 7 T. All 172 participants were scanned at 3 T with the 1.25 mm isotropic dMRI data HCP acquisition protocol ($b = 0, 1000, 2000, 3000 \text{ s/mm}^2$)³¹ as well as at 7 T with the 1 mm isotropic dMRI data HCP acquisition protocol ($b = 0, 1000, 2000 \text{ s/mm}^2$).⁴³ Tractograms were reconstructed with Tractoflow and as detailed above for both 3 T and 7 T datasets, with the same parameters. Therefore, although the acquisition protocols are slightly different, they provide high resolution, high number of diffusion directions, high signal-to-noise ratio and, most importantly, provide a great test-retest paradigm to evaluate the Arnold tracts reconstruction (reproducibility results are shown in the [Supplementary material](#)).

We thus created a single volumetric 3D probabilistic template for each of the Arnold tracts by averaging all 172 3 T and 7 T left and right Arnold tracts in a separate 3D NIFTI volumes. These volumes represent the probability that a given voxel belongs to each Arnold tract in the population. In parallel, we also created streamline templates, as described in Chenot et al.⁴⁴ In short, for each bundle, we concatenated all subjects' streamlines to achieve a population representation of each bundle. We then removed spurious streamlines⁴⁵ with unique shape that were not coherent with the overall population. As we had no explicit reason to expect inter-hemispheric differences, we opted for a symmetric template to avoid injecting a hemispheric bias. We thus simply mirrored each bundle left/right and concatenated the results. These steps resulted in dense population symmetric templates for each of the Arnold tracts using streamlines (rather than voxels). Symmetric templates are useful for algorithms such as Recobundles⁴⁶ or BST.³³

Finally, OR and ILS/IFOF were similarly reconstructed, to analyse the spatial relationships between these tracts and Arnold tracts.

Post-mortem Klingler fibre dissection

Twelve cerebral hemispheres obtained from the body donation programme of the University of Tours were used to study the pulvino-temporal projections and their anatomical relationships.

We used a variant of the method described by Ludwig and Klingler, in which the specimens were fixed through bilateral carotid injection of formalin, a procedure that has been demonstrated to be effective in previous studies of our team.^{47–49} They were then kept in formalin for at least 2 months, frozen twice at -20°C for 14 days and then slowly defrosted.

Two approaches were used for microdissection: dissection from the ventral surface and dissection from the temporal isthmus. In dissection from the ventral surface, after studying the sulcogyral anatomy, microdissection consisted of removing the cortical grey matter, made brittle by the preparation, preserving the firmer sub-cortical white matter lamina, which contains short intergyral association fibres ('U-fibres'). This procedure was performed along the entire length between the collateral and the occipitotemporal sulcus, a region that partially corresponds to the floor of the temporal horn of the lateral ventricle. Progressive dissection of the same region exposes the corresponding segment of the basal longitudinal system (also called inferior longitudinal fasciculus). The ventricular floor is then opened through a small window starting at the level of the collateral sulcus and medial portion of the fusiform gyrus. It is then enlarged to broadly expose the ventricular cavity ependyma, choroid plexus and hippocampus. The latter, as well as part of the uncus and parahippocampal gyrus, are then removed. Dissection of fibres from the ventricular roof and adjacent areas was performed in layers after the ablation of the ependyma. Two approaches were used to expose the fibre contingent directed to the temporal operculum: from a dissection of the medial superior surface of the temporal operculum after ablation of the extreme capsule; from previously sectioned hemispheres in an oblique coronal slice plane to remove the occipital lobe, most of the parietal lobe, and to obtain posterior visualization of the pulvinar, metathalamus and temporal peduncle.

For the exploration of the white matter, fibre bundles were identified and followed as much as possible toward their terminations. Delicate customized wooden spatulas and microsurgical instruments were used for fibre dissection. Dissection on the ventricular roof, adjacent areas and temporal isthmus was performed using an M655 surgical microscope (Wild Heerbrugg Co.), whenever the naked eye was no longer sufficient. The results were photographically documented and presented along with the tractography results in similar perspectives, for comparison.

Results

Clinical cases

We report intraoperative findings in four patients (three females, one male) operated on under awake conditions for a left insular isocitrate dehydrogenase (IDH)-mutated grade II glioma. The median age was 40 years (range 34–44). Seizures were the presenting symptom in all cases (generalized seizures in two cases and partial seizures in two cases). The median tumour preoperative volume was 30 cm^3 (range 20–45). The median postoperative residual volume was 1.9 cm^3 (range 1–3.2). All patients returned to a normal socio-professional life 4 months after surgery.

A complete description of the cortical mappings is given in the [Supplementary Fig. 2](#). Regarding white matter mapping (complete description [Supplementary Table 1](#)), an anomia site was found in all four patients in the temporal isthmus, at the medio-postero-superior corner of the resection (red dots in [Supplementary Fig. 3](#)). Stimulation intensity was set at 2 mA in Patients 1, 2 and 4, and 3 mA in Patient 3. In tractography dissections, the ILS was not found

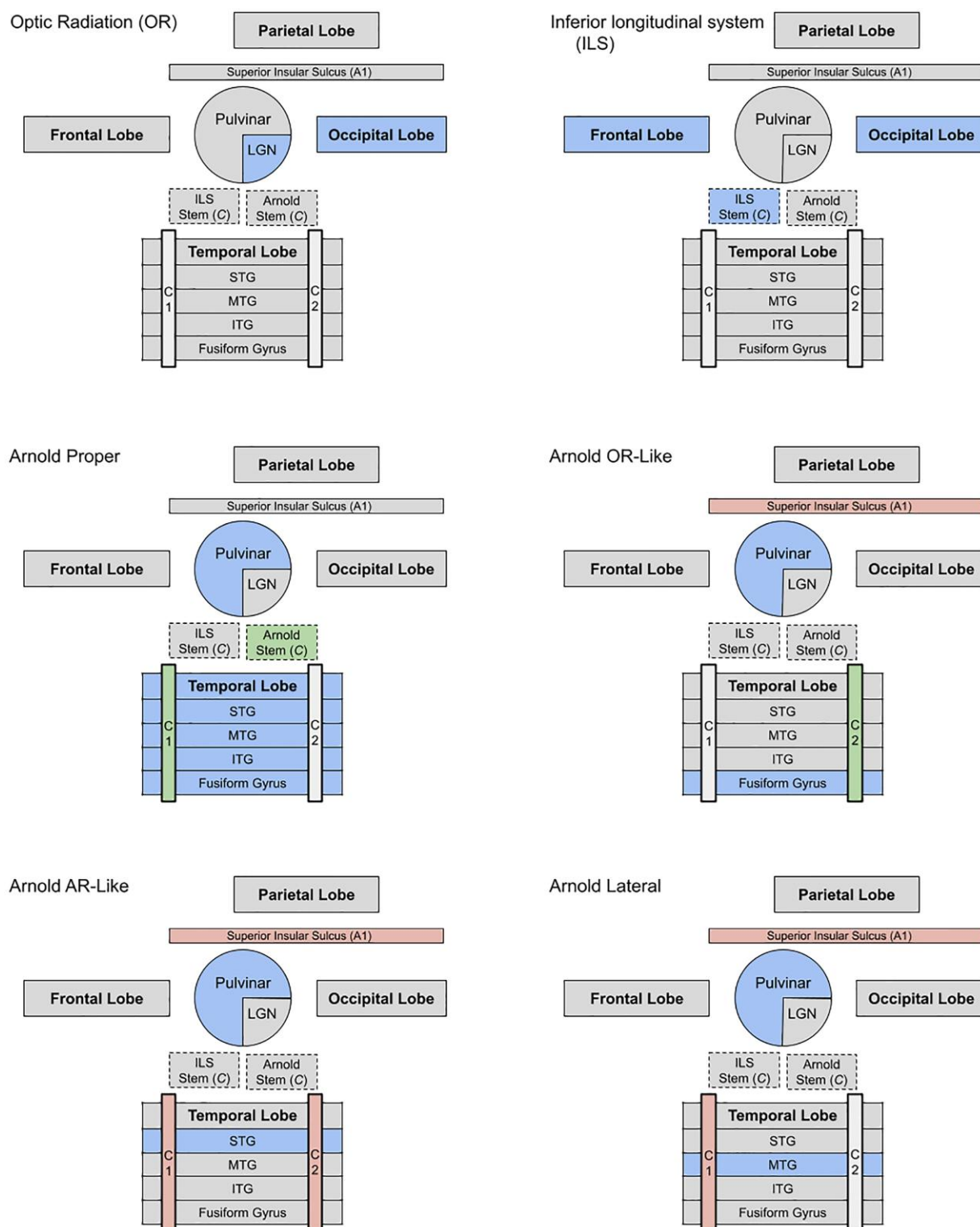


Figure 1 Schematic representation of dissections rules. Boxes represent cortical and white matter areas. Blue colour corresponds to the rule 'either end', green colour to 'any part' and red colour to 'NOT any part'. The following rules were applied: for the optic radiations: LGN 'either end' AND occipital lobe 'either end'; for the ILS/IFOF: frontal lobe 'either end' AND 'either end' AND ILS stem 'any part'; for Arnold proper: temporal lobe 'either end' AND pulvinar 'either end' AND Arnold stem 'any part' AND C1 'any part'; for Arnold OR-like: fusiform gyrus 'either end' AND pulvinar 'either end' AND C2 'any part' AND A1 'any part not'; for Arnold AR-like: superior temporal gyrus 'either end' AND pulvinar 'either end' AND C1 'any part not' AND C2 'any part not' AND A1 'any part not'. AR = auditory radiations; IFOF = inferior-fronto-occipital fasciculus; ILS = inferior longitudinal system; ITG = inferior temporal gyrus; LGN = lateral geniculate nucleus; MTG = middle temporal gyrus; OR = optic radiations; STG = superior temporal gyrus.

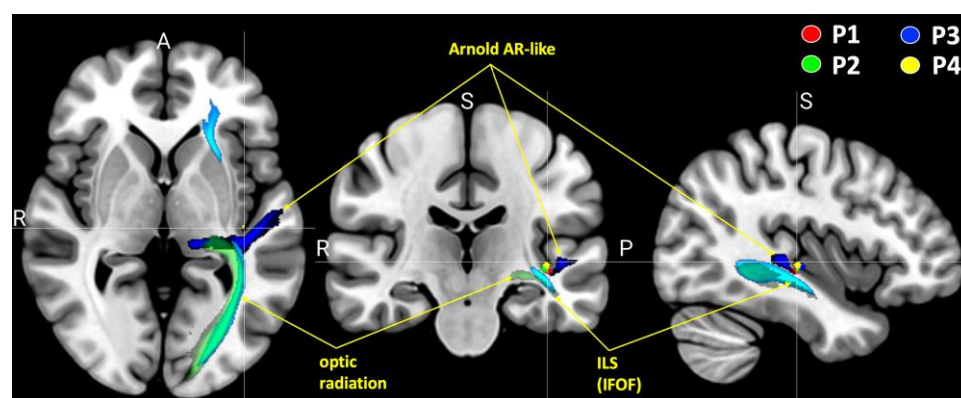


Figure 2 Anomia sites in relation to template tracts. The anomia sites were located in the auditory radiations (AR)-like component of pulvino-temporal connections, except for Patient 3. In this case, the site was located slightly more anteriorly, in keeping with a greater intensity of stimulation (3 mA instead of 2 mA). IFOF = inferior-fronto-occipital fasciculus; ILS = inferior longitudinal system; OR = optic radiations; P1–P4 = Patients 1–4; P = posterior; R = right; S = superior.

at all in three patients, while scant streamlines could be found in Patient 2 (Supplementary Fig. 3). To complement individual tractography, we also overlaid the ILS streamline template to the postoperative T₁-weighted image. The ILS was found to project entirely in the resection cavity in Patients 2 and 3, while part of it bordered the resection cavity in Patients 1 and 4. In all cases, the AR-like component of Arnold tracts was reconstructed successfully by tractography and the stimulation point was touching it, except in Patient 3, in whom the stimulation was close to it. Interestingly, in the two cases (Patients 1 and 3) in whom cortical mapping identified an anomia site in the superior temporal gyrus, the sites matched the cortical endings of the AR-like component. Figure 2 shows the four stimulation points in the MNI template, overlaid with the ILS, OR and Arnold AR-like tracts obtained from healthy subjects (see below).

Finally, acute postoperative language evaluations, performed within 6 days after surgery, demonstrated phonological troubles in the four patients (Supplementary Table 2), in addition to the expected typical triad of the ventral stream dysfunction (anomia, semantic paraphasia, verbal perseverations). Such phonological troubles are supposed to be supported by the dorsal route: pars opercularis/ventral premotor cortex—third branch of superior longitudinal fasciculus and arcuate fasciculus—posterior temporal areas, none of which were touched by the resection. We thus attribute this impairment to the oedema surrounding the functional boundary of the AR-like tract. The fact that phonological disturbances fully resolved at 4 months (contrary to the ventral stream dysfunctions, Supplementary Table 2) gives further support to the hypothesis that these troubles were transiently generated by acute postoperative oedema surrounding the responsive sites in the AR-like tract.

Tractography in healthy subjects

Tractography made it possible to study the anatomical relationships of the different components of pulvino-temporal radiations among themselves and between these components and the two prominent neighbouring bundles that travel inside the temporal stem: the ILS/IFOF and the OR (Fig. 3 and Supplementary Fig. 4). The deepest tract was the Arnold proper component. It was overlaid by the OR and the OR-like component. Intermingled within this layer, we found the streamlines of the lateral component. Streamlines of the ILS adopted a superior situation in relation to the OR and the lateral and OR-like components. The AR-like component crossed the ILS just superior to it.

The test-retest HCP data at 3 T and matching subjects at 7 T successfully reconstructed all four Arnold tracts. The most stable and less variable tracts were the Arnold proper and AR-like tracts. The OR-like and Arnold lateral tracts were harder to track due to their shape, high curving section and high overlap with other nearby white matter bundles. Despite these challenges, the templates reconstructed at 3 T versus 7 T were highly reproducible. The quantitative results (Supplementary Tables 3–5) collectively demonstrated the robustness and reproducibility of the tractography methods used in the study. Intra-subject analysis using both 3 T and 7 T modalities revealed Dice coefficients ranging from 0.173 to 0.584, and bundle adjacency ranging from 0.683 to 2.595 mm. These findings highlight the precision of individual tractography reconstruction, with bundle adjacency values notably below the 2 mm threshold, indicating good intra-subject consistency.

When comparing individual subjects to the template, the range of Dice coefficients was 0.111 to 0.293 across 3 T and 7 T templates, with bundle adjacency spanning from 1.338 to 2.596 mm. The relatively higher bundle adjacency values reflect the expected anatomical variability at the population level, with values still falling below the 4 mm threshold, considered very good for population-level assessments. The inter-template comparison between 3 T and 7 T modalities showed Dice coefficients between 0.737 and 0.862, demonstrating a high degree of consistency in capturing anatomical variability at the population level. The bundle adjacency values ranged from 0.171 to 0.42 mm, underscoring the reliability of these tractography methods across different scanners.

Fibre dissection and anatomical description

Figures 4–7 detail the correspondences between observations in the dissection of tractograms and specimens.

The anterior pulvino-temporal fasciculus (Arnold's bundle proper) was present in all dissected specimens with a high level of similarity in terms of general morphology but also in terms of origins and terminations (Fig. 4). A triangular-shaped bundle at the posterior apex starts from the lateral portion of the pulvinar and travels anteriorly medially to the loop of Meyer, partially covered by the tail of the caudate nucleus, anteriorly reaching the temporal pole where it opens in a fan.

An infero-lateral contingent was present in all specimens dissected to that effect (Fig. 5). Its trajectory runs side-by-side with the OR, hence its name OR-like component. In dissection, the fibres

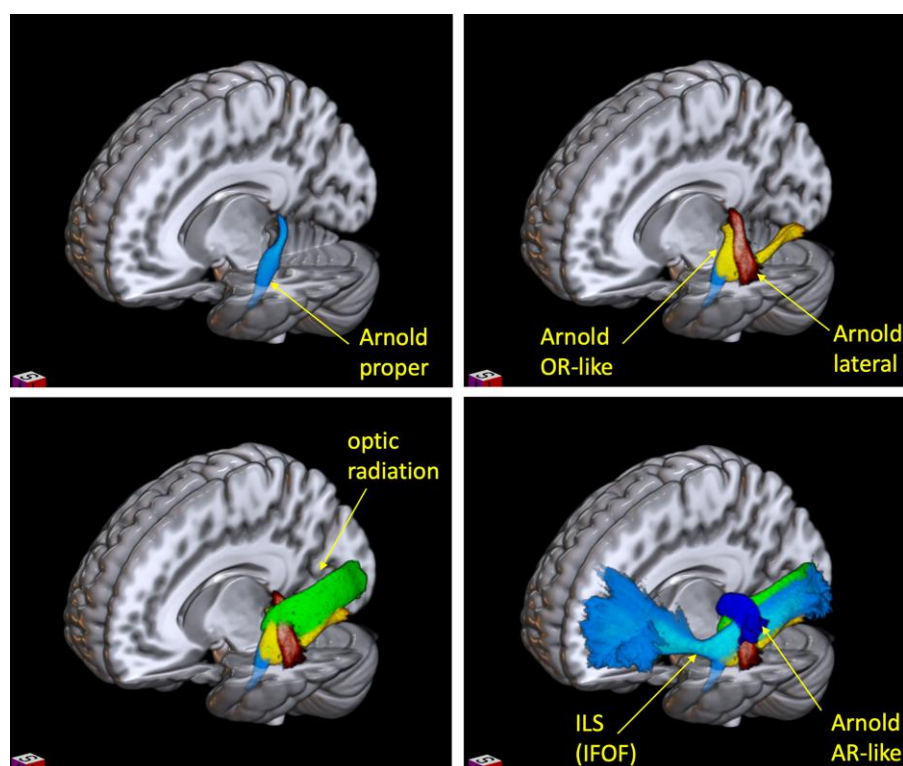


Figure 3 Step-by-step virtual dissection of the different tracts. IFOF = inferior-fronto-occipital fasciculus; ILS = inferior longitudinal system; AR = auditory radiations; IFOF = inferior-fronto-occipital fasciculus; ILS = inferior longitudinal system; OR = optic radiations.

that emerge from the pulvinar accompany the others from the posterior thalamic radiations, especially those from the OR originating from the lateral geniculate body (LGB), and with them penetrate the Meyer's loop, the most posterior being covered medially and superiorly by the tapetum that runs under the ependymal surface. Fibres that leave the ventricular wall to join the lateral face of the temporal lobe could not be followed in microdissection. The presence of the sagittal stratum, a large contingent of highly predominant horizontal fibres does not allow us to completely rule out nor affirm that part of fibres originating from the pulvinar extends to the lateral or inferior aspect of the temporal lobe. However, tractogram dissections reproducibly identified the posterior fusiform gyrus as the cortical target of these connections.

A third lateral component was directed towards the middle temporal gyrus (Fig. 6). On tractography, most fibres followed a relatively straight trajectory, ending in the mid-part of the middle (and inferior) temporal gyrus, while some rare fibres curved posteriorly in the core white matter of the temporal lobe, ending in the posterior part of the middle temporal gyrus. These findings were in good agreement with dissections, although fibres could not be followed until their cortical target.

The fourth contingent directed supero-laterally, towards the temporal operculum and superior temporal gyrus, was evidenced with a remarkable agreement with tractography (Fig. 7). These fibres were closely related to the AR, which penetrates the thickness of the superior temporal gyrus originating from the medial geniculate body. We thus referred to these connections as the AR-like component. We observed with both techniques that these fibres were directed towards the planum temporale, just posterior to the lateral part of the gyrus of Heschl.

Discussion

Historical review of pulvino-temporal connections

An important point of discussion regarding the pulvino-temporal connections involves the possible reasons why they have been relatively sidelined in anatomical studies thus far. Different possible factors were identified in our study. The first—historical—relates to the fact that the main studies related to the topic were published in the pre-war and inter-war period only in German, which may have hampered their dissemination. To the best of our knowledge, there has been no publication or translation to English language of the remarkable works of Arnold, Probst (or even Klingler) on this specific topic,^{25,29,50,51} limiting their scientific impact in the second half of the 20th century.

Another—methodological—factor is the fact that significant fibre crossing exists along the trajectory of these connections. This renders their dissection laborious and delicate, even with the advent of Klingler's technique.⁵² Such difficulties may have discouraged the topic study.

Finally, as demonstrated in the present study, these bundles go along with other well defined elements of thalamic radiation, such as OR and AR.^{25,29,50,51,53,54} This continuity seems to have induced a relative reductionism through which the identity of adjacent (non-optical, non-auditory) fibres aroused less interest.

We now propose to review historical anatomical studies, comparing our results with previous work on fibre dissection.

Arnold proper

In his work *Handbuch der Anatomie des Menschen: mit besonderer Rücksicht auf Physiologie und praktische Medizin*, Arnold describes

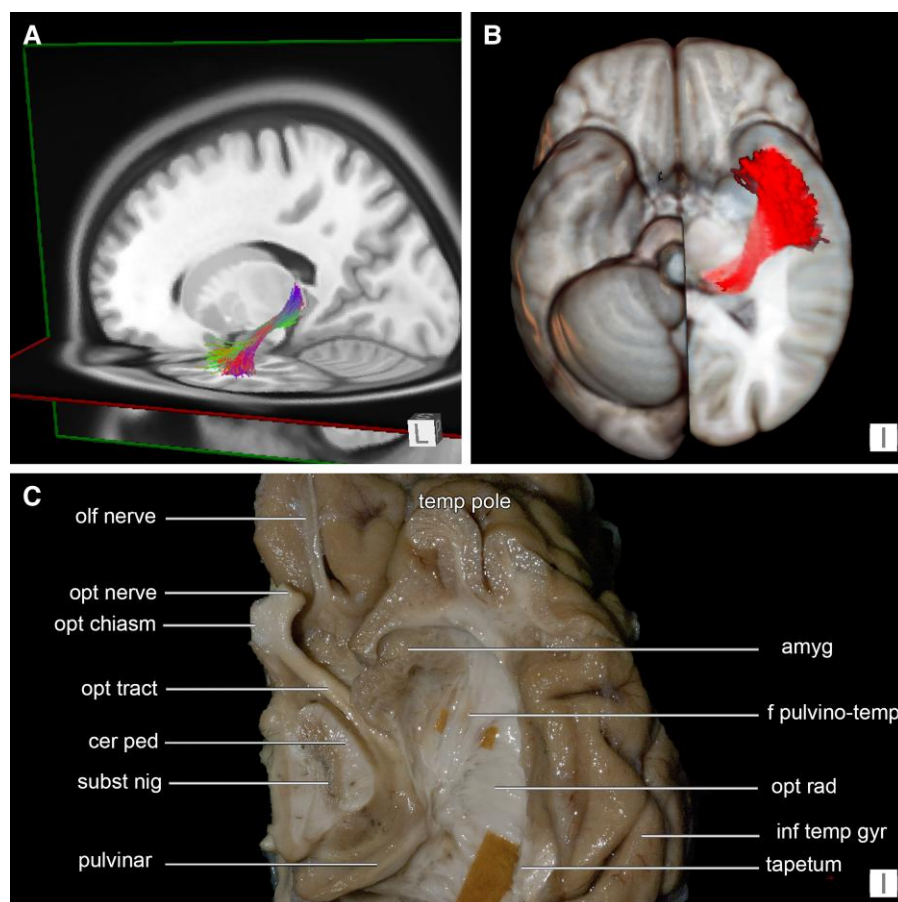


Figure 4 Anterior component of pulvino-temporal radiations (Arnold's bundle proper). (A) Tractography in the left cerebral hemisphere based on 20 Human Connectome Project (HCP) subjects imaged at 7 T, supero-lateral view. (B) Volume rendering reconstruction, inferior view. (C) Result of fibre dissection according to Klingler's technique. amyg = amygdala; cer ped = cerebral peduncle; f pulvino-temp = fasciculus pulvino-temporalis (anterior component of pulvino-temporal radiations, Arnold's bundle proper); inf temp gyr = inferior temporal gyrus; olf = olfactory; opt = optic; opt rad = optic radiations; subst nig = substantia nigra.

fibres located in the lower portion of the 'Stabkranzes' ('bar wreath', corona radiata) directed to the 'inferior lobe' (temporal lobe) in a section dedicated to the structure of the cerebral hemisphere.⁵⁰ Illustrations documenting his observations appear in the first issue of his *Tabulae anatomicae*, published some years earlier (Fig. 8A and B).²⁵ This work did not contain any detailed description but identified this white matter pathway—this time in Latin—as '*Fibrae corona radiatae quae ad lobi inferioris apicem pertinent*' (fibres of the corona radiata that extend to the apex of the inferior lobe). This fibre bundle was then referred to by both J. Dejerine and G. Roussy in their works.^{55,56}

Arnold optic radiations-like

The Arnold fasciculus was revisited in more detail by Ranson in 1921, in a study that detailed other components of the sublenticular portion of the internal capsule and how it contributes to the formation of the roof of the temporal horn of the lateral ventricle.⁵⁷ His important contribution was to recognize that the interconnections between the posterior thalamus and the temporal lobe were not just those described by Arnold. He noticed that another set of fibres, compact and partially covered by the terminal stria, started from the 'stratum zonale', on the dorsal surface of the posterior portion of the thalamus, followed the tail of the caudate nucleus, and

spread laterally over the roof of the temporal horn (Fig. 8C). This lateral extension was the main difference from the previous bundle described by Arnold. Ranson named this second contingent as the 'fasciculus thalamo-temporalis arcuatus'.⁵⁷

Most classical descriptions of posterior thalamic radiation mention the temporal (Meyer's) loop that surrounds the ventricular temporal horn as being composed solely of fibres from the geniculocalcarine tract. From the end of the 20th century, several authors challenged this view, demonstrating that the geniculocalcarine fibres surround the ventricular cavity amid a more complex system of thalamic projections. As noted by Türe et al.,⁵⁴ OR is accompanied by a great number of pulvinocortical fibres. In the posterior thalamic peduncle, the transition between the fibres that emerge from the LGB and those that emerge from the pulvinar is progressive and harmonious.

Detailed analysis of posterior thalamic radiations, therefore, shows that they contain much more than projections to the calcarine cortex and that the structure macroscopically identified as Meyer's loop is not formed exclusively by fibres destined for it. Goga and Türe⁵³ addressed this issue in a study in which the temporal loop was exposed from both the lateral and lower aspects of the cerebral hemisphere. The authors draw attention to the fact that it is not possible to safely individualize fibres belonging to optical radiation from the other accompanying thalamic radiations.

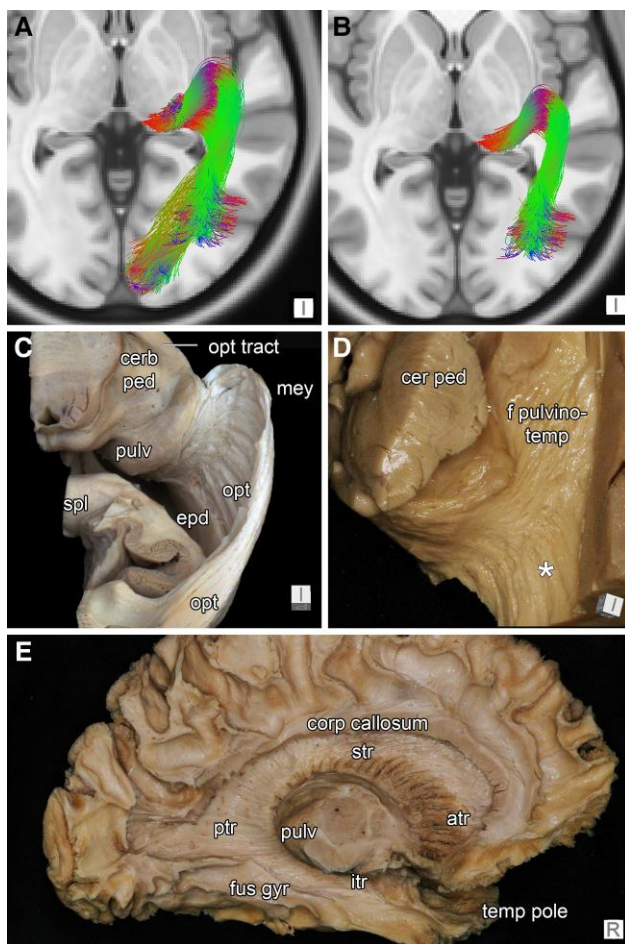


Figure 5 Optic radiation (OR)-like component of pulvino-temporal radiations. (A) Tractography in the left cerebral hemisphere based on 20 Human Connectome Project (HCP) subjects imaged at 7 T, inferior view. (B) Volume rendering reconstruction, inferior view. (C–E) Result of fibre dissection according to the Klingler's technique of the optic radiation (C), OR-like component (D, asterisk) and an overview of thalamic radiations (E). The thick (dominant) contingent of horizontal fibres in the sagittal stratum did not allow us to affirm the cortical terminations of the OR-like component of pulvino-temporal radiations using fibre dissection. Tractogram dissections reproducibly identified the posterior fusiform gyrus as the cortical target of these pathways. atr = anterior thalamic radiation; cerb ped = cerebral peduncle; corp = corpus; epd = ependyma; f pulvino-temp = fasciculus pulvino-temporalis (anterior component of pulvino-temporal radiations, Arnold's bundle proper); fus gyr = fusiform gyrus; itr = inferior thalamic radiation; mey = Meyer's loop; opt = optic; ptr = posterior thalamic radiation; pulv = pulvinar; spl = splenium of corpus callosum; str = superior thalamic radiation.

They also emphasized that, according to their observations, all projection fibres of the sublenticular internal capsula that reach the sagittal stratum seem to do so through the temporal loop over the temporal horn.

Arnold auditory radiations-like

Ranson's observations resembled those of Probst carried out on non-human primates using an experimental injury method and published—such as Arnold—only in German.⁵¹ However, in addition to the fibres surrounding the ventricle, Probst noticed that some were directed superiorly, running in the external and extreme capsules and reaching the superior temporal gyrus. He reports

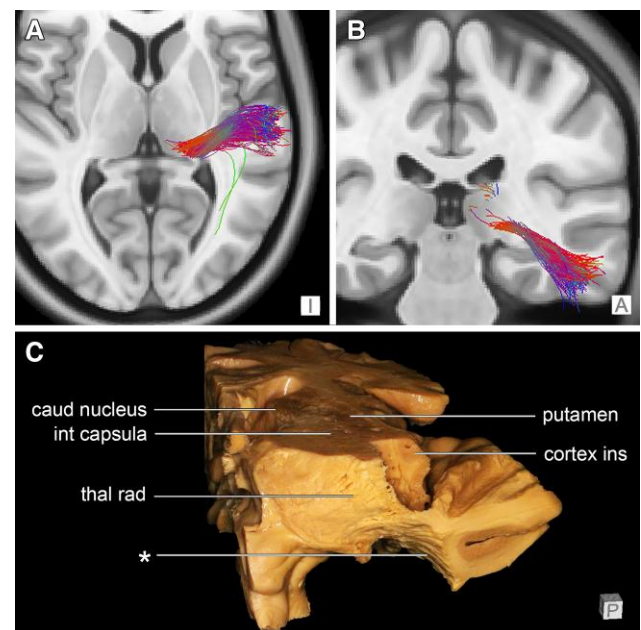


Figure 6 Lateral component of pulvino-temporal radiations. (A and B) Tractography in the left cerebral hemisphere based on 19 Human Connectome Project (HCP) subjects imaged at 3 T, axial (A) and coronal (B) view. (B) Volume rendering reconstruction, inferior view. (C) Result of fibre dissection according to Klingler's technique. The thick (dominant) contingent of horizontal fibres in the temporal isthmus did not allow us to determine the cortical terminations of the lateral component (asterisk) of pulvino-temporal radiations using fibre dissection. Tractogram dissections reproducibly identified the middle temporal gyrus as the cortical target of these pathways. caud = caudate; ins = insula; thal rad = thalamic radiation.

having also noticed them in some microcephalic brains, in which these fibres were particularly evident for some reason.^{51,57,58}

Arnold lateral

Dissection studies underwent important methodological advances from the 1950s onwards, after the seminal works of Ludwig and Klingler.^{52,59,60} Penfield's attention has been turned to thalamic connections to the temporal lobe in the 1950s.²⁹ He was motivated by Smyth and Stern's observation of a thalamic tumour inducing aphasia.⁶¹ Penfield wondered if language troubles engendered by cortical stimulation might not be due, at least in part, to a thalamic functional interference through thalamocortical projections. At Penfield's request, Joseph Klingler performed dissections of the thalamic radiations. The authors reported that laboratory observations were in favour of the existence of such connections running over the temporal horn. To the best of our knowledge, one single photograph of a relatively coarse dissection accompanied by an anatomical drawing (Fig. 8D) has been published in Penfield and Roberts' book *Speech and Brain Mechanism*.^{29(pp218,219)} This depiction fits well with the present lateral component.

Comparison with previous connectivity studies in humans

MRI-based structural connectivity

Few studies have attempted to describe Arnold's fasciculus with dMRI.^{62–65} All have exclusively mentioned the Arnold proper component. In his recent atlas of the temporal lobe connectivity,

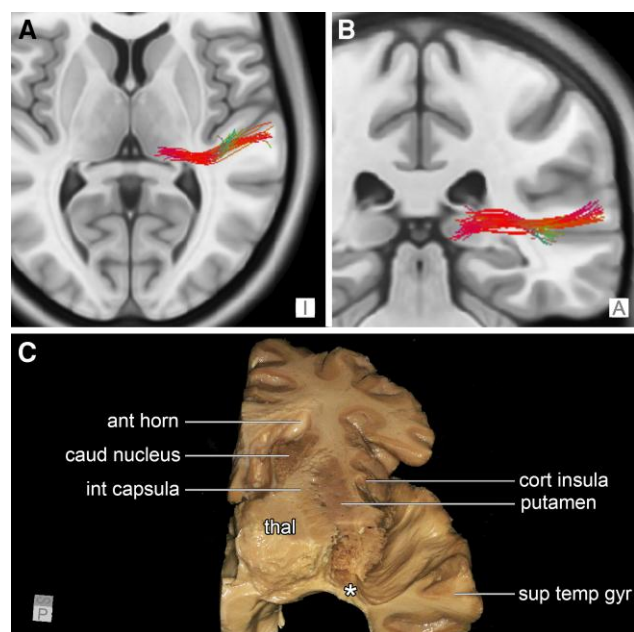


Figure 7 Auditory radiation (AR)-like component of pulvino-temporal radiations. These fibres run close to the auditory radiations, penetrate the planum temporale and terminate posterior to the anterior temporal (Heschl's) gyrus. (A and B) Tractography in the left cerebral hemisphere based on 20 Human Connectome Project (HCP) subjects imaged at 7 T, axial (A) and coronal (B) view. (C) AR-like component of pulvino-temporal radiations (asterisk) as evidenced by fibre dissection according to the Klingler's technique. ant = anterior; caud = caudate; cort = cortical; int = nterior; sup temp gyr = superior temporal gyrus; thal = thalamus.

Catani⁶⁶ briefly described thalamo-temporal connections, distinguishing a component originating in the lateral pulvinar—running parallel to the auditory radiations—and a component originating in the medial pulvinar, without depicting the trajectory nor the cortical targets of these two components. Hence, the present work is the first to provide an in-depth tractography analysis of the structural anatomy of pulvino-temporal connections.

Thanks to our HCP 3 T/7 T test-retest analysis on 172 participants, we provide the first—to the best of our knowledge—template of the Arnold tracts, both in the format of a NIFTI volumetric probabilistic 3D map and in the TRK format, as a streamline template in MNI space. We also demonstrated (Supplementary Tables 3–5) that the reconstruction of Arnold tracts are robust and accurate.

MRI-based functional connectivity

The pulvinar has regularly appeared to be functionally connected to the temporal lobe in functional imaging studies.^{67–69} There seems to be a good agreement between the present structural connectivity findings and previous work on functional connectivity of the pulvinar.⁶⁷ The locations of areas AT (anterior temporal), FFA/FBA (fusiform face area/fusiform body area), pSTS (posterior superior temporal sulcus), latTemp (lateral temporal) could fit with the topography of the cortical endings of the four Arnold's components found in the present study, proper, OR-like, AR-like and lateral, respectively. However, it should be acknowledged that our study could not determine the distinct subparts of the pulvinar in which the different branches could find their origin. This limitation precludes to compare accurately our results with those previously mentioned.

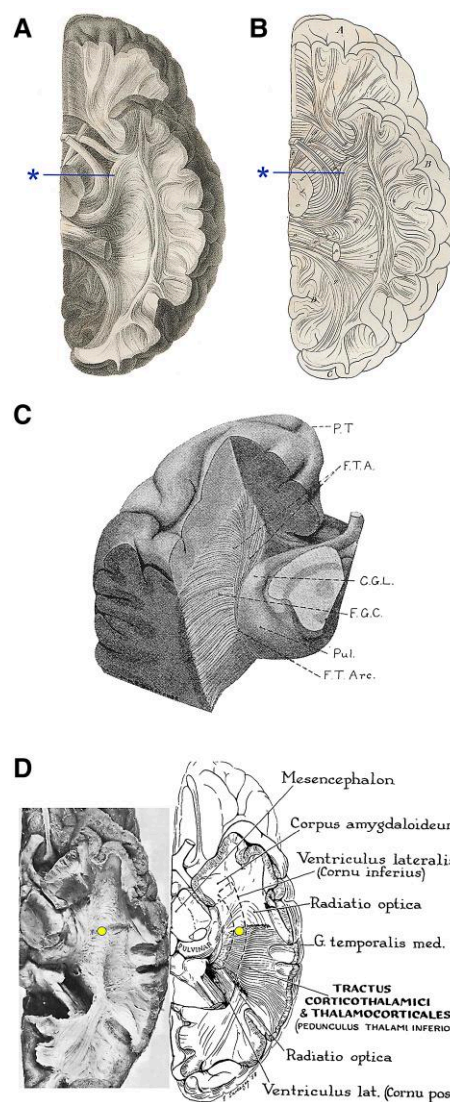


Figure 8 Pulvino-temporal connections according to classical anatomical studies in the 19th and 20th centuries. (A and B) Anterior pulvino-temporal radiations (asterisk) according to Arnold.²⁵ (C) Anterior and optic radiation-like components, as described and named F.T.A. by Ranson.⁵⁷ (D) Lateral component (yellow dot) described by Penfield²⁹ based on an anatomical preparation from Ludwig Klingler.

Electrophysiologically-based effective connectivity

One study investigated the distribution of cortical evoked potentials when stimulating the medial pulvinar with single pulse, in the context of patients implanted with depth electrode for exploration of drug-resistant epilepsy.⁷⁰ Although there is a sparse sampling of the temporal lobe, the data confirmed strong connectivity with the entire temporal lobe, including the regions corresponding to the terminations of the four components.

Implications for networks of picture naming

We have demonstrated here that stimulation of pulvino-temporal connections in the temporal isthmus causes anomia, establishing for the first time, their causal role in picture naming. Our observations are in agreement with previous work reporting anomia when stimulating the left dominant pulvinar.⁷¹ It is tempting to integrate

these results in the framework of a cortico-striato-thalamo-cortical model of picture naming recently proposed by our group.⁷² In this model, the striatum is receiving, from the different cortical areas of the lexico-semantic system, converging inputs that include all lexico-semantic items activated by a single picture (e.g. the picture of a tree automatically activates several items such as ‘tree’, ‘branches’, ‘wood’, ‘green’, ‘leaves’, etc.). The striato-thalamic system acts as an adjustable filter, which selects the best answer according to the context. The thalamic feedback then amplifies the appropriate response (and/or inhibits the competitors) in the different cortical areas. Such a model has been demonstrated to be in agreement with the observation of verbal perseverations when stimulating the striatal inputs coming from the lexico-semantic system⁷²: in such condition, the state of the thalamus fails to be updated and the response to previous items is re-sent to the cortex. This model would align with lexical access viewed as a procedural memory process, thought to be supported by frontal areas (inferior and superior frontal gyrus), dorsal caudate nucleus and the ventral anterior and anterior part of ventral lateral nuclei of the thalamus.¹⁶ In cooperation with this procedural memory, a parallel declarative process has been proposed¹⁶ supposedly supported by the (left) pulvinar and its connection with the (left) inferior frontal gyrus. In this framework, the existence of four branches of pulvino-temporal connections would help to bind together the different levels of representation of a selected item: the OR-like component would act on visual word-form⁷³ and visual object-form⁷⁴ coding (as it projects to the mid-posterior fusiform gyrus), the lateral component would act on an amodal lexical level (as it projects close to basal language temporal areas) (see also Cohen *et al.*⁷⁵), the AR-like component would act at the level of the lexico-phonological interface (as it projects close to Wernicke area, see Patients 1 and 3), and the Arnold proper component would act at the conceptual level (as it projects to the temporal pole, where concepts are thought to be represented^{76,77}). This multilevel integration would also fit with the multilevel model of the arcuate fasciculus recently proposed⁷⁸ and would facilitate the binding of a unique item at all of its levels of representation.

Limitations

Our results are subject to the usual limitations of tractography and Klingler’s dissections in regions with major crossing of fibres. An acknowledged challenge for tractography concerns bottleneck regions, in which one fibre orientation distribution function (FODF) exists in the voxel, but many different fibre populations are known to go through it.⁷⁹ Hands-on dissections suffer the same caveat. Hence, as the techniques for studying structural connectivity continue to evolve, it is possible that the presented systematization will appear to still be reductionist and other trajectories for pulvino-temporal pathways or other connections such as temporo-putaminal inputs might also exist. On the contrary, both techniques are also subject to false positives, and part of the identified connections might have been artificially reconstructed.

Finally, there is some uncertainty pertaining to the manual localization of the stimulation points in the postoperative MRI. However, the fact that the stimulation was applied at the medial-postero-superior corner of the postoperative cavity strongly reduces the risk of mislocalization.

It should also be acknowledged that we do not have any data regarding the behavioural correlates of the other branches than the AR-like connections. More studies are needed to explore the neuropsychological consequences—if any—of damage specific to each of the three other branches.

Conclusion

In this paper, we have demonstrated morphological, radiological and functional aspects of pulvino-temporal connections. The results obtained with contemporary study techniques—diffusion MRI tractography, operative microscope-assisted fibre dissection and white matter stimulation in awake surgery—are in good agreement with historical, but disremembered anatomical and functional works, as well as with lesion studies. We have described the presence of at least four fundamental fibre contingents: the anterior component, corresponding to the pulvino-temporal fasciculus initially described by Arnold in the 19th century and destined to the anterior temporal lobe; the OR-like component, which leaves the pulvinar accompanying the optical radiations and reaches the posterior basal temporal cortices; the lateral component, which crosses orthogonally the temporal stem, towards the middle (and inferior) temporal gyrus, and the AR-like component, which leaves the pulvinar and accompanies the auditory radiations to the superomedial aspect of the temporal operculum. The observation that electrical stimulation in the AR-like component induces anomia establishes the role of these connections in the framework of cortico-thalamo-cortical loops sustaining the lexical process of picture naming.

Data availability

We have created an open-source repository at https://github.com/scilus/atlas_arnold. This repository contains:

- (i) The probabilistic template.
 - (a) The MNI reference 3D nifti volume used
 - (b) A 3D nifti volume per Arnold tract
- (ii) The streamline representation of the template.
 - (a) A trk file per Arnold tract
- (iii) Bash script describing how a tractogram can be virtually dissected in all the Arnold tracts

Includes all ROIs necessary in MNI space.

MRI data used in the preparation of this work were obtained from the MGH-USC Human Connectome Project (HCP) database (<https://ida.loni.usc.edu/login.jsp>). The automated script for tracking Arnold tracts is available.

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Competing interests

The authors report no competing interests.

Supplementary material

Supplementary material is available at *Brain* online.

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